The Journal of Open Source Software

GroupFunctions.jl: computing individual entries of the irreducible representations of the unitary group U(d)

David Amaro-Alcalá [1] **and Konrad Szymański** [1]

1 Research Centre for Quantum Information, Institute of Physics, Slovak Academy of Sciences, Dúbravská cesta 9, Bratislava, Slovakia

DOI: N/A

Software

- Review
- Repository
- Archive

Editor: Open Journals

Reviewers:

- @openjournals

Submitted: 01 January 1970
Published: 01 January 1970

Summary

`GroupFunctions.jl`[1] is a Julia library for computing individual matrix elements of irreducible representations of $\mathrm{U}(d)$. These matrix elements, called group functions, can be evaluated symbolically or numerically. For $\mathrm{SU}(2)$, they reduce to the Wigner D-functions. The library computes these matrix elements in a carrier-space basis enumerated by Gelfand-Tsetlin patterns (Gelfand & Tsetlin, 1950). It can also compute entire representation operators, construct input unitaries from parameterisations common in quantum optics, translate Gelfand-Tsetlin patterns into occupation-number kets, and compute the associated Schur functions.

Statement of need

Representations of the unitary group $\mathrm{U}(d)$ arise in many subfields of physics and mathematics, and computations often reduce to evaluating their matrix elements, called group functions. For an irrep labelled by λ, the corresponding object is

$$D^{(\lambda)}_{\mathrm{out},\mathrm{init}}(U) = \langle \mathrm{out} \mid D^{(\lambda)}(U) \mid \mathrm{init} \rangle,$$

where init and out denote basis states in the representation carrier space.

In mathematics, summing the diagonal group functions gives the trace of the representation matrix of U. This trace is the character of the representation and the Schur polynomial of the eigenvalues of U, an object central to algebraic combinatorics and symmetric function theory.

In quantum physics, group functions appear in several settings. In quantum optics, a group function gives the transition amplitude of photons through a linear optical network (de Guise et al., 2014). The same object can be used to characterise the performance of quantum devices (Amaro-Alcalá, 2026) and to describe the symmetry properties of states (Otto & Szymański, 2024). One important subproblem is boson sampling (Aaronson & Arkhipov, 2011), where the transition amplitude reduces to a permanent whose evaluation is classically computationally hard. Whether noisy real-world quantum devices can perform this computation remains an open question.

Some of these tasks require only a numerical estimate, whereas others require an exact symbolic group function. `GroupFunctions.jl` addresses the latter. Although related software exists (see below), to our knowledge none is designed to compute a single representation-matrix entry as an exact symbolic D-function. We therefore compare the cost of computing one such

[1] David Amaro-Alcalá developed the package, its algorithms, and its test suite, and prepared the initial documentation. Konrad Szymański substantially revised and expanded the documentation and developed additional examples, and contributed to the standardisation of the API.

entry with the conventional route that constructs the Lie-algebra representation matrices and exponentiates the resulting full matrix.

Let $\lambda \vdash N$, with $N > 1$, and let the compatible basis states A, B be supplied directly. For fixed λ, A, B and $d \to \infty$, the hook–content formula gives

$$D_\lambda(d) = \frac{f^\lambda}{N!} d^N + O(d^{N-1}) = \Theta(d^N).$$

The present implementation evaluates one symbolic D-function using $\Theta(d^3)$ structural operations and $\Theta(d^2)$ peak storage. By contrast, even if construction of the $D_\lambda(d) \times D_\lambda(d)$ Lie-algebra matrix is granted at no cost, conventional unstructured scaling-and-squaring/Padé exponentiation with classical dense arithmetic requires $\Theta(D_\lambda(d)^3) = \Theta(d^{3N})$ operations and $\Theta(D_\lambda(d)^2) = \Theta(d^{2N})$ storage. Consequently, the direct-to-exponentiation ratios are

$$\Theta\big(d^{-3(N-1)}\big) \quad \text{and} \quad \Theta\big(d^{-2(N-1)}\big),$$

respectively. Thus, computing one requested D-function has an asymptotic advantage over constructing the complete representation matrix in this fixed-N, growing-d regime.

State of the field

Several existing packages relate to `GroupFunctions.jl` but address different problems. `SUNRepresentations.jl` (QuantumKitHub contributors, 2020) and the algorithm presented in the appendix of (Alex et al., 2011) compute $\mathrm{SU}(d)$ Clebsch-Gordan coefficients. `RepLAB` (Rosset et al., 2021) supports manipulating irreducible representations of various groups, including $\mathrm{U}(d)$, but provides only indirect numerical access to group functions. `IntegrateUnitary.jl` (Pawela & Puchała, 2026) performs symbolic integration over compact groups rather than evaluating representation matrices. `haarpy` (Cardin et al., 2024) implements Weingarten-calculus methods. Other libraries focus on quantum optics. `BosonSampling.jl` (Seron & Restivo, 2024), `Perceval` (Heurtel et al., 2023), and `QOptCraft` (Aguado et al., 2023) numerically model linear optical devices, while `The Walrus` (Gupt et al., 2019) helps compute amplitudes for Gaussian boson sampling. These packages do not target the symbolic computation of individual representation-matrix entries, which is the primary purpose of `GroupFunctions.jl`.

Software design

This library provides a unified method for computing representation matrix elements of $\mathrm{U}(d)$ irreps specified by integer partitions of length at most d, including computations with symbolic input matrices. Several computational routes are possible in principle. For symmetric irreps, which model fully indistinguishable bosons, one can manipulate states as polynomials of creation operators applied to the vacuum. Another approach constructs and exponentiates the Lie algebra generators in the chosen representation. Both approaches are computationally expensive. More restricted methods based on generating functions also exist (Prakash, 1996).

The authors chose the Grabmeier-Kerber formula (Grabmeier & Kerber, 1987) as the most general solution. It expresses the matrix element as a sum of monomials in the entries of the input matrix, weighted by the irreducible representation and the states in question. Our implementation optimises the enumeration of the double cosets that index this sum by grouping permutations that contribute the same monomial. This implementation provides the library's main function, `group_function`.

Internally, the algorithm represents basis states as semistandard Young tableaux. The user-facing functions expose the equivalent Gelfand-Tsetlin patterns through the `GTPattern` data structure and provide utility functions for common quantum optics scenarios, such as `occupation_number`.

Research impact statement

`GroupFunctions.jl` has been used to evaluate matrix elements and group characters in published work on filtered randomized benchmarking (Amaro-Alcalá, 2026). The library has been under continuous development since 2020 and is tested in CI; it is registered in the Julia General registry under the MIT licence, installable with the following command:

```
] add GroupFunctions
```

The following example evaluates, symbolically, a matrix element between states of the $\mathrm{U}(4)$ mixed-symmetry irrep.

```julia
λ = [2,2,1,0]; basis=basis_states(λ); # integer partition and basis
group_function(λ, basis[1], basis[end]) # symbolic matrix element
```

The result is a polynomial in the entries of the $\mathrm{U}(4)$ matrix. For symmetric irreps the matrix element reduces to a permanent, which symbolic algebra packages also compute; for mixed-symmetry ones no existing software computes these elements symbolically. Further examples, applications, and mathematical background are available in the documentation.

AI usage disclosure

OpenAI Codex (GPT-5.3) assisted with optimising the performance of the double-coset enumeration at a late stage. The authors developed the mathematical design and proof of correctness of the optimised algorithm. Anthropic Claude (Opus 4.8) assisted with code review and language review of the documentation and manuscript. The authors reviewed and edited all AI-assisted changes.

Acknowledgements

We thank Dr. Hubert de Guise for helpful discussions and suggestions on the bibliography, and Dr. Alonso Botero for suggestions that improved the presentation. Mitacs CALAREO, DeQHOST APVV-22-0570, QUAS VEGA 2/0164/25, Postdokgrant APD0161, and the Stefan Schwarz programme supported this work. David Amaro-Alcalá acknowledges the indirect support of the Government of Alberta and NSERC during his PhD studies at the University of Calgary.

References

Aaronson, S., & Arkhipov, A. (2011). The computational complexity of linear optics. *Proceedings of the Forty-Third Annual ACM Symposium on Theory of Computing*, 333–342. https://doi.org/10.1145/1993636.1993682

Aguado, D. G., Gimeno, V., Moyano-Fernández, J. J., & Garcia-Escartin, J. C. (2023). QOptCraft: A Python package for the design and study of linear optical quantum systems. *Computer Physics Communications*. https://doi.org/10.1016/j.cpc.2022.108511

Alex, A., Kalus, M., Huckleberry, A., & Delft, J. von. (2011). A numerical algorithm for the explicit calculation of SU(N) and SL(N,C) Clebsch-Gordan coefficients. *Journal of Mathematical Physics*, *52*(2), 023507. https://doi.org/10.1063/1.3521562

Amaro-Alcalá, D. (2026). Kostant relation in filtered randomized benchmarking for passive bosonic devices. In *Physical Review A* (Vol. 113). American Physical Society. https://doi.org/10.1103/xmlq-vssg

Cardin, Y., de Guise, H., & Quesada, N. (2024). Haarpy, a Python library for Weingarten calculus and integration of classical compact groups and ensembles. In *GitHub repository* (0.0.6). https://github.com/polyquantique/haarpy; GitHub.

de Guise, H., Tan, S.-H., Poulin, I. P., & Sanders, B. C. (2014). Coincidence landscapes for three-channel linear optical networks. *Physical Review A*, *89*(6). https://doi.org/10.1103/physreva.89.063819

Gelfand, I. M., & Tsetlin, M. L. (1950). Finite-dimensional representations of the group of unimodular matrices. *Doklady Akademii Nauk SSSR*, *71*, 825–828.

Grabmeier, J., & Kerber, A. (1987). The evaluation of irreducible polynomial representations of the general linear groups and of the unitary groups over fields of characteristic 0. *Acta Applicandae Mathematicae*, *8*(3), 271–291. https://doi.org/10.1007/bf00046717

Gupt, B., Izaac, J., & Quesada, N. (2019). The Walrus: A library for the calculation of hafnians, Hermite polynomials and Gaussian boson sampling. *Journal of Open Source Software*, *4*(44), 1705. https://doi.org/10.21105/joss.01705

Heurtel, N., Fyrillas, A., Gliniasty, G. de, & others. (2023). Perceval: A software platform for discrete variable photonic quantum computing. *Quantum*, *7*, 931. https://doi.org/10.22331/q-2023-02-21-931

Otto, F., & Szymański, K. (2024). Rotational covariance restricts available quantum states. *Physical Review A*, *110*(2), 022440. https://doi.org/10.1103/PhysRevA.110.022440

Pawela, Ł., & Puchała, Z. (2026). *IntegrateUnitary.jl: A Julia package for symbolic integration over Haar measures*. https://arxiv.org/abs/2605.23830

Prakash, J. (1996). Wigner's D-matrix elements for SU(3)-a generating function approach. *Journal of Physics A: Mathematical and General*, *29*(18), 6059–6072. https://doi.org/10.1088/0305-4470/29/18/032

QuantumKitHub contributors. (2020). *SUNRepresentations.jl*. https://github.com/QuantumKitHub/SUNRepresentations.jl.

Rosset, D., Montealegre-Mora, F., & Bancal, J.-D. (2021). *RepLAB: A computational/numerical approach to representation theory*. https://github.com/replab/replab; Springer International Publishing. https://doi.org/10.1007/978-3-030-55777-5_60

Seron, B., & Restivo, A. (2024). BosonSampling.jl: A Julia package for quantum multi-photon interferometry. *Quantum*, *8*, 1378. https://doi.org/10.22331/q-2024-06-18-1378